

Title - Metabolism

Liam is sprinting away from a bull.

Describe and explain **three** ways his body will respond to the demands of sprinting. (3 marks)

1. Increased heart rate to pump blood more quickly to deliver oxygen and glucose needed for respiration in muscles cells so they can contract.
2. Increased breathing rate to increase the amount of oxygen absorbed by the blood and meet demand for extra oxygen in muscle cells.
3. Increased breathing volume to increase the amount of oxygen absorbed in each breath to meet demand for extra oxygen in muscle cells.

Metabolism

Learning objectives:

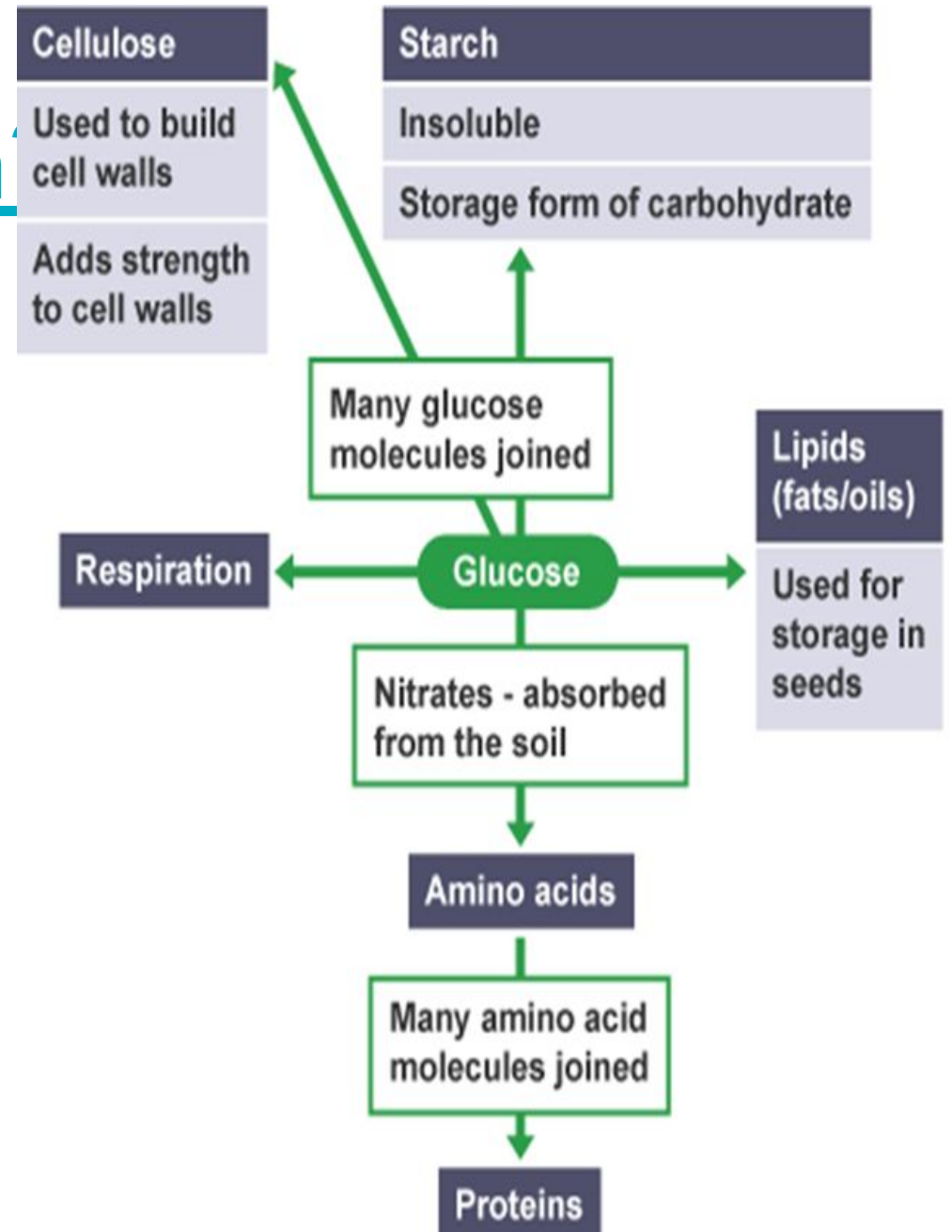
By the end of the lesson you will be able to:

- Define the term metabolism
- Explain the importance of sugars, amino acids, fatty acids and glycerol in the synthesis and breakdown of carbohydrates, proteins and lipids.
- Give examples of reactions in metabolism
- HT – Describe how the liver is involved in repaying the oxygen debt

Key words:
Metabolism
Starch
Glycogen
Cellulose
Lipid
Proteins

What is metabolism

- People normally think of metabolism as how easily their body burns off food
- **Metabolism** is the term used for all the chemical reactions that go on inside an organism's body.
- It is the sum of all the reactions that take place in the cell or in the body

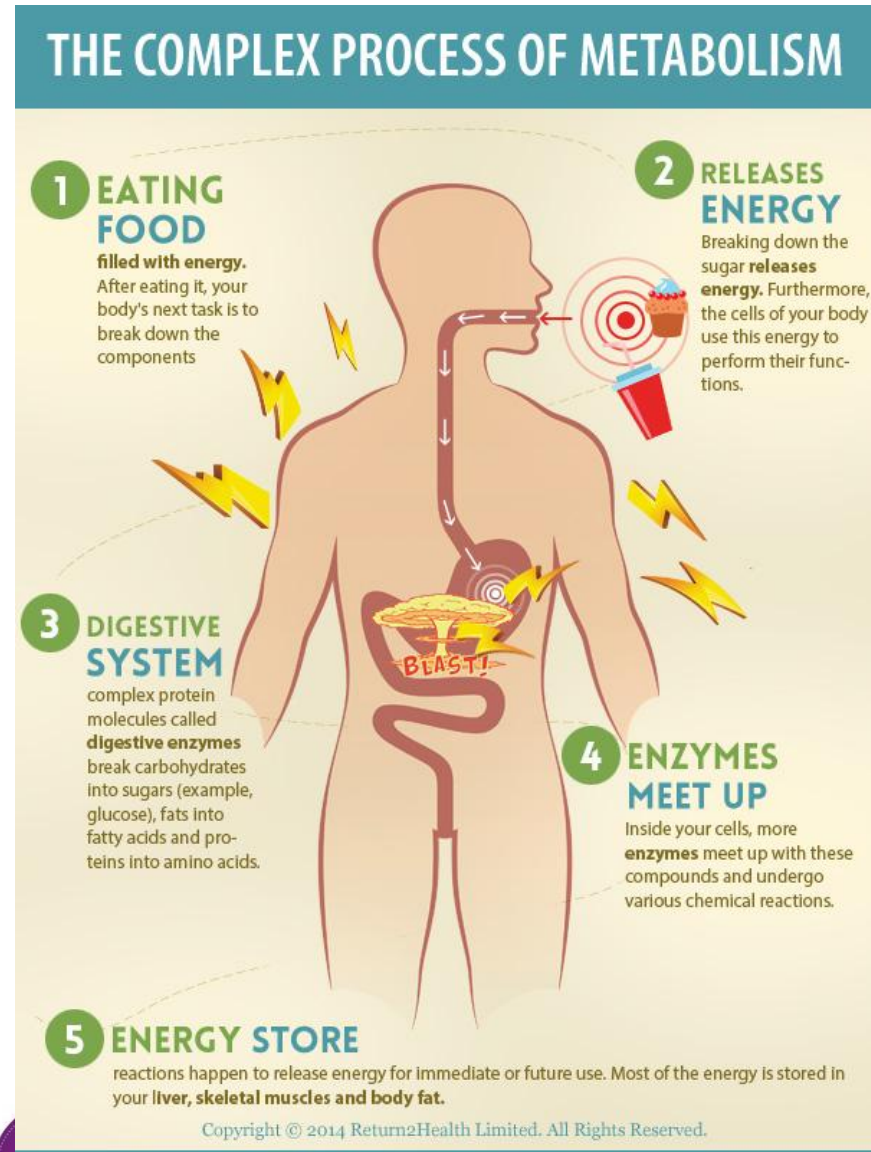


List as many metabolic reactions you can.

1:59

Examples Of Metabolic Reactions write down 3

- Conversion glucose into glycogen, cellulose, starch
- Formation of lipid (fat) from fatty acids and glycerol
- Use of glucose and nitrates to form amino acids then proteins (plants only)
- Respiration
- Photosynthesis (plants only)
- Breakdown of excess protein in the liver to form urea (urine)





Metabolism is the **sum of all the chemical reactions** in a cell or the body.



ATION

Exercise is just as important as diet. Why?



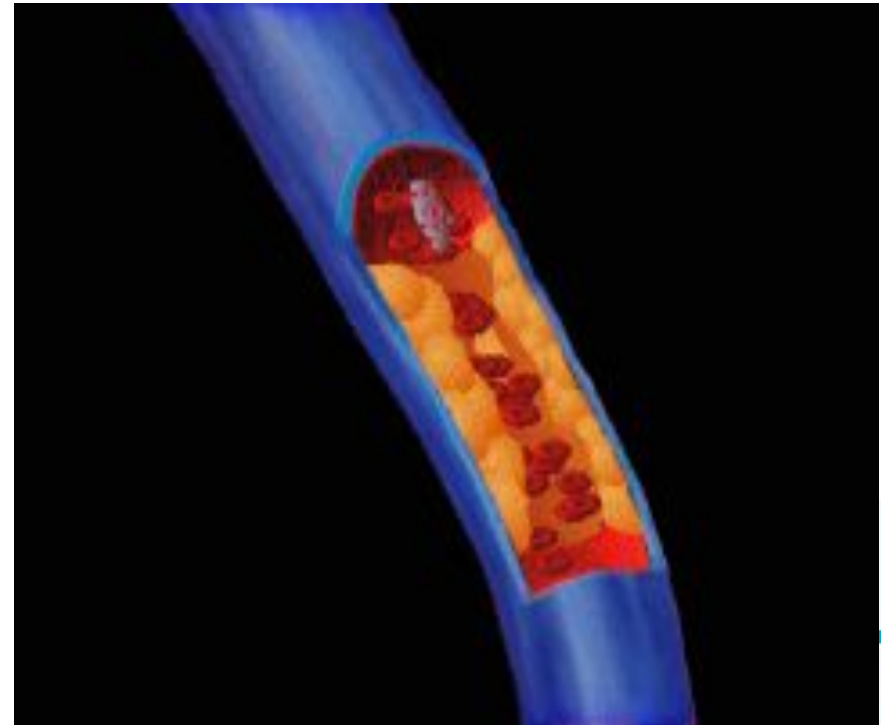
Why is exercise important?

- Increases the amount of energy used by the body
- Decreases the amount stored as fat
- Builds muscle and boosts metabolic rate
- The more energy you use (and less fat you store), the less likely you are to become obese!



TOO MUCH FOOD AND TOO LITTLE EXERCISE CAN LEAD TO...

- Arthritis – worn joints
- Type 2 Diabetes – high blood sugar level
- High blood pressure
- Heart and blood vessel disease (cardiovascular)

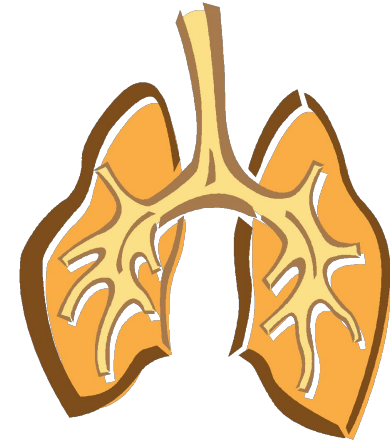




Why do
people's energy
needs vary?



If there are more reactions taking place (for example, respiration) then we need more energy! So there will be a higher metabolic rate.

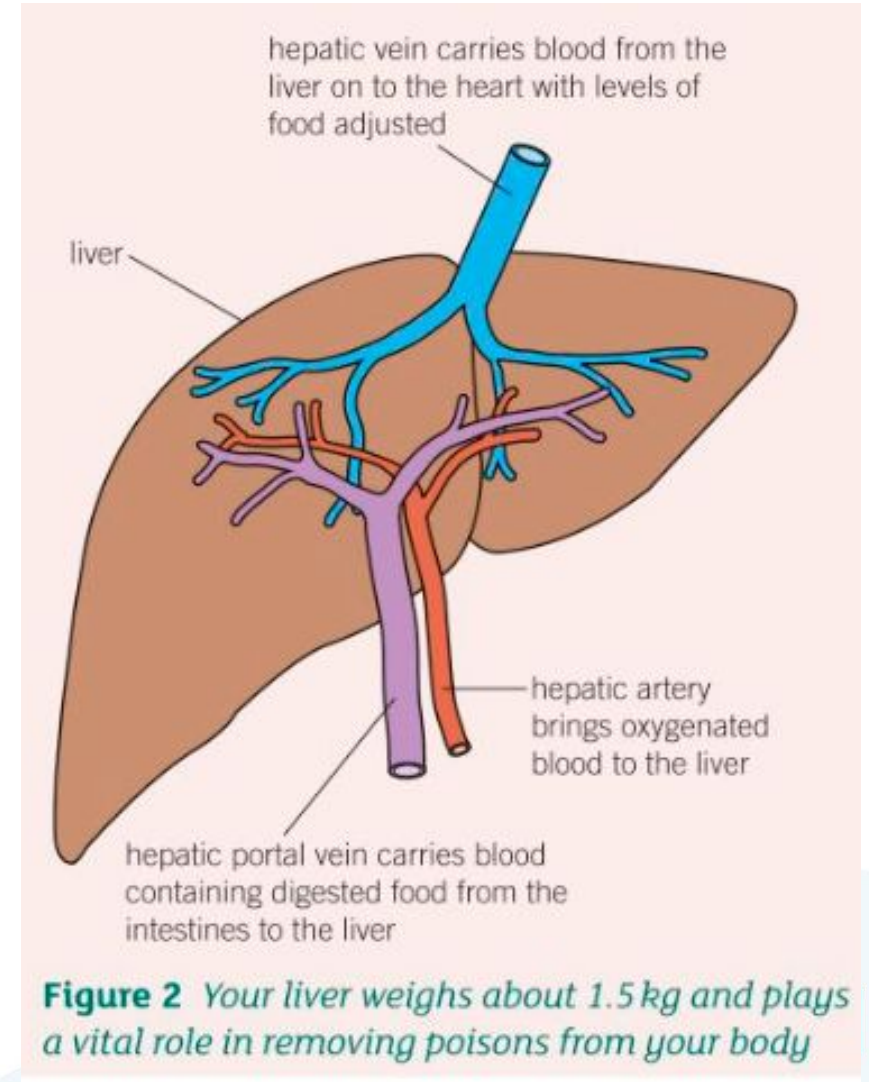


The liver

- What is the primary role of the liver?
- The produce bile to help break down fat
- The liver also plays other roles:
 1. Detoxifying the blood- Detoxification
 2. Storing energy (excess glucose) as glycogen- Glucose regulation
 3. Breaking down excess amino acids- Deamination

Higher Tier Information – The Role of The Liver

- The liver is heavily involved with many different metabolic functions, such as:
 - Detoxifying poisonous substances such as ethanol from alcoholic drinks
 - Passing the breakdown products into the blood so they can be excreted in the urine via the kidneys
 - Breaking down old, worn out blood cell and storing the iron until it is needed to synthesise more blood cells.
 - Dealing with lactic acid produced by anaerobic respiration.



Plenary

1. What factors could effect the metabolic rate? (2 marks)
2. Lactic acid production exercise affects an athlete's performance. Explain why lactic acid is produced during exercise. (2 marks)



Exam style question

b. Draw a line to match the smaller molecules with the larger molecules.

Glucose

Fatty acids and glycerol

Amino acids

Lipids

Proteins

Carbohydrates

[3 marks]

Plenary

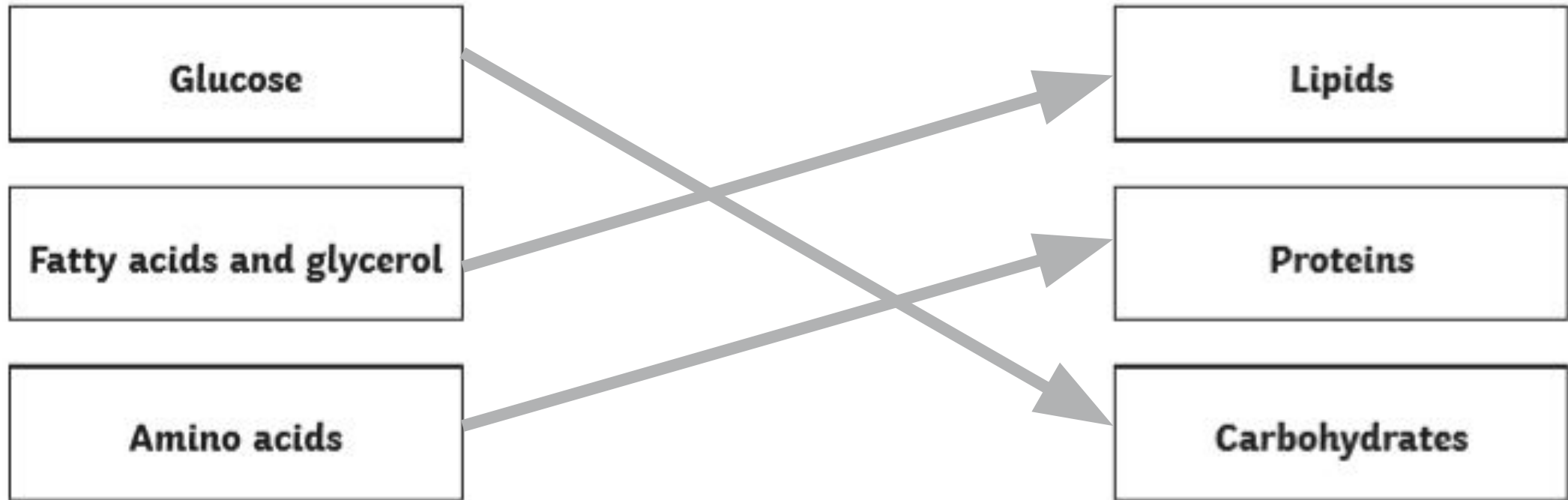
1. What factors could effect the metabolic rate? (2 marks)

Answer - Age | Gender | Inherited Factors |
Body Size/Weight | Exercise Levels |
Proportion of Muscle to fat | Pregnancy

2. Lactic acid production exercise affects an athlete's performance. Explain why lactic acid is produced during exercise. (2 marks)

Answer – Insufficient/no oxygen available (1 mark), for (just aerobic respiration/respire anaerobically (1 mark).

b. Draw a line to match the smaller molecules with the larger molecules.



[3 marks]

Homework

Log on to BBC Bitesize/Kerboodle
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DO NOW

Make Notes and complete the